Dividing fractions

1 Simplify:

a $\frac{1}{8} \div \frac{1}{5}$ b $\frac{1}{2} \div \frac{9}{10}$ c $\frac{1}{5} \div \frac{3}{7}$ d $\frac{3}{4} \div \frac{1}{9}$

e $\frac{1}{10} \div \frac{1}{3}$ f $\frac{1}{2} \div \frac{1}{9}$ g $\frac{2}{3} \div \frac{1}{8}$ h $\frac{1}{11} \div \frac{3}{7}$

2 Simplify:

a $\frac{2}{7} \div \frac{5}{3}$ b $\frac{9}{10} \div \frac{10}{9}$ c $\frac{2}{7} \div \frac{7}{18}$ d $\frac{3}{15} \div \frac{4}{3}$

e $\frac{28}{9} \div \frac{4}{9}$ f $\frac{3}{15} \div \frac{15}{4}$ g $\frac{21}{27} \div \frac{3}{2}$ h $\frac{5}{7} \div \frac{45}{2}$

i $\frac{2}{3} \div \frac{11}{15}$ j $\frac{10}{7} \div \frac{10}{7}$ k $\frac{4}{5} \div \frac{36}{35}$ l $\frac{3}{7} \div \frac{14}{27}$

m $\frac{2}{36} \div \frac{4}{7}$ n $\frac{3}{55} \div \frac{7}{20}$ o $\frac{10}{99} \div \frac{45}{22}$ p $\frac{22}{27} \div \frac{3}{2}$

3 Simplify:

a $3 \div \frac{1}{11}$ b $\frac{1}{9} \div 8$ c $\frac{4}{3} \div 5$ d $\frac{6}{11} \div 8$

e $\frac{3}{5} \div 4$ f $\frac{65}{9} \div 5$ g $3 \div \frac{39}{10}$ h $7 \div \frac{98}{12}$

4 Simplify:

a $1\frac{2}{5} \div \frac{3}{4}$ b $2\frac{1}{4} \div 1\frac{2}{5}$ c $1\frac{2}{7} \div 3\frac{1}{2}$ d $3\frac{1}{3} \div 2\frac{1}{2}$

e $2\frac{1}{3} \div 6\frac{2}{12}$ f $3\frac{3}{8} \div 2\frac{1}{4}$ g $6\frac{3}{4} \div 4\frac{2}{7}$ h $8\frac{1}{2} \div 1\frac{5}{6}$

5 Simplify:

a $\frac{1}{5} \times \frac{1}{3} \div \frac{1}{2}$ b $\frac{2}{3} \div \frac{1}{9} \times \frac{2}{5}$ c $\frac{3}{6} \div \frac{12}{5} \div \frac{1}{2}$ d $\frac{3}{4} \div \frac{2}{5} \times \frac{1}{6}$

e $\frac{1}{7} \times \frac{2}{9} \div \frac{4}{27}$ f $\frac{1}{7} \times \frac{5}{8} \div \frac{15}{49}$ g $\frac{18}{25} \div \frac{9}{10} \times \frac{1}{3}$ h $\frac{8}{11} \div \frac{2}{3} \times \frac{15}{2}$

Applications

6 A family drinks $\frac{3}{4}$ of a carton of milk per day. How many days will 6 cartons of milk last the family?

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7 A glass holds $\frac{3}{8}$ of a litre of lemonade. How many glasses of lemonade can be filled with 6 litres of lemonade?

8 Yang eats $\frac{2}{5}$ of a packet of bicuits each day. If he buys 12 packets, how many days will the packets last?

9 If $\frac{2}{3}$ of a tart is shared equally between 5 people, what fraction of the tart would each person receive?

10 A truck tank contains $42\ \text{L}$ when $\frac{3}{4}$ full. What is the capacity of the tank when full?

11 If a garden snail can move $\frac{1}{100}\ \text{km/h}$, how many hours would it take for the garden snail to reach $\frac{3}{4}\ \text{km}$?

12 Christine has $1\frac{3}{4}$ of a metre of ribbon. She uses $\frac{7}{20}$ to wrap a present. How many presents can she wrap with her ribbon?

13 Sheldon is planning to sell banana bread on the upcoming school fair. If a loaf requires $\frac{3}{4}$ cups of sugar, how many banana bread loaves can Sheldon make if he has $8\frac{2}{8}$ cups of sugar on hand?

14 Jean is making bows for her gift wrapping business. A bow requires $6\frac{1}{6}$ inches of ribbon. If she has $55\frac{1}{2}$ inches of ribbon, how many bows can Jean make?

15 Benjamin wants to engrave his name on a metal plate. There is $1\frac{2}{4}\ \text{cm}$ space on the plate. There are three sizes of letters to choose from: $\frac{1}{8}\ \text{cm}$ wide, $\frac{1}{6}\ \text{cm}$ wide and $\frac{1}{4}\ \text{cm}$ wide.

a Find the maximum number of letters that Benjamin can engrave on the plate using the following sizes:

i $\frac{1}{8}\ \text{cm}$ wide ii $\frac{1}{6}\ \text{cm}$ wide iii $\frac{1}{4}\ \text{cm}$ wide

b Which letter sizes could he use for his name?

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